

Condition	Size	Mean	Median	SD	SE	Min	Max	p-value
Novelty Score								
Human Ideas	49	4.86	5.00	1.26	0.18	1.50	7.00	–
AI Ideas	49	5.62	5.50	1.39	0.20	1.50	8.33	0.03*
AI Ideas + Human Rerank	49	5.78	6.00	1.07	0.15	3.00	8.33	0.00**
Excitement Score								
Human Ideas	49	4.56	4.33	1.16	0.17	2.00	7.00	–
AI Ideas	49	5.18	5.50	1.33	0.19	2.50	7.33	0.08
AI Ideas + Human Rerank	49	5.45	5.50	1.36	0.19	1.00	7.33	0.00**
Feasibility Score								
Human Ideas	49	6.53	7.00	1.50	0.21	3.00	9.00	–
AI Ideas	49	6.30	6.00	1.27	0.18	2.50	8.50	1.00
AI Ideas + Human Rerank	49	6.41	6.50	1.06	0.15	4.00	9.00	1.00
Expected Effectiveness Score								
Human Ideas	49	5.10	5.33	1.14	0.16	3.00	7.00	–
AI Ideas	49	5.48	5.50	1.23	0.18	2.00	7.50	0.58
AI Ideas + Human Rerank	49	5.57	5.50	0.99	0.14	3.00	7.50	0.17
Overall Score								
Human Ideas	49	4.69	4.67	1.16	0.17	2.00	6.67	–
AI Ideas	49	4.83	5.00	1.34	0.19	1.50	7.50	1.00
AI Ideas + Human Rerank	49	5.32	5.50	1.24	0.18	2.00	7.50	0.06

Table 8: Scores across all conditions by averaging the scores for each idea and treating each idea as one data point (Test 2). Size is the number of ideas for each condition, and the p-values are computed with two-tailed Welch’s t-tests with Bonferroni correction. We **bold** results that are statistically significant (* $p < 0.05$; ** $p < 0.01$). AI ideas are judged as significantly better than human ideas in terms of novelty while being comparable on all other metrics.

3 where we treat each reviewer as one datapoint and compute their average score on each condition. Then for each reviewer, we get their mean score difference between the AI Ideas condition and the Human Ideas condition, as well as the difference between the AI Ideas + Human Rerank condition and the Human Ideas condition. This way, we only analyze the differences among the different conditions. That is, if the differences are significantly higher than zero under the one-sample t-test, that indicates reviewers are giving higher scores to one condition compared to the other. The results are shown in Table 9, and we see significant results ($p < 0.05$) that AI ideas in both the AI Ideas and AI Ideas + Human Rerank conditions are rated more novel than Human Ideas. Therefore, we conclude that AI ideas generated by our ideation agent are judged as more novel than human expert generated ideas, consistently across all three different statistical tests.¹⁰

6 In-Depth Analysis of the Human Study

While the above main results highlight the promise of LLMs in generating novel research ideas, there are some additional nuances. In this section, we move beyond the statistical comparisons and dive

¹⁰We also include results of fitting linear mixed-effects models in Appendix N, which reinforces our conclusions. Additionally, we plot the breakdown of all metrics by topic in Appendix O.

	N	Mean Diff	p-value
Novelty Score			
AI Ideas vs Human Ideas	70	0.94	0.00**
AI Ideas + Human Rerank vs Human Ideas	65	0.86	0.00**
Excitement Score			
AI Ideas vs Human Ideas	70	0.73	0.01*
AI Ideas + Human Rerank vs Human Ideas	65	0.87	0.00**
Feasibility Score			
AI Ideas vs Human Ideas	70	-0.29	0.36
AI Ideas + Human Rerank vs Human Ideas	65	-0.08	0.74
Effectiveness Score			
AI Ideas vs Human Ideas	70	0.42	0.16
AI Ideas + Human Rerank vs Human Ideas	65	0.39	0.16
Overall Score			
AI Ideas vs Human Ideas	70	0.24	0.36
AI Ideas + Human Rerank vs Human Ideas	65	0.66	0.01*

Table 9: Mean score differences between AI ideas and human ideas by treating each reviewer as a data point (Test 3). All p-values are computed with one-sample t-tests with Bonferroni correction. We **bold** results that are statistically significant (* $p < 0.05$; ** $p < 0.01$).

into other aspects of our collected data. Specifically, we focus on the quality of human ideas, reviewer preferences, and the extent of reviewer agreement.

6.1 Human Experts May Not Be Giving Their Best Ideas

We first investigate whether human experts are submitting their best ideas to us. We did a post-study survey to understand how idea-writing participants came up with their ideas. Out of the 49 participants, 37 of them came up with the idea on the spot, while the other 12 already had the idea before the study. Furthermore, we asked the survey question: “How does this idea compare to your past research ideas (ideas that you actually worked on)? Please answer with a percentile. E.g., this idea is one of my top 10% ideas.” Our participants indicated that on average their submitted ideas are about the top 43% of all their past ideas. This implies that our collected ideas are likely the median-level ideas from these expert researchers, which is reasonable given that most of them came up with the idea within the 10-day time constraint of the task.

6.2 Reviewers Tend to Focus More on Novelty and Excitement

To gain a deeper understanding of the dynamics between the different metrics in the review process, we explore whether reviewers focus on specific aspects when evaluating the ideas. We compute the pairwise correlation between different metrics in Table 10. The overall score mostly correlates with the novelty score ($r = 0.725$) and excitement score ($r = 0.854$), while having almost no correlation ($r < 0.1$) with the feasibility score. This implies that reviewers might be paying more attention to the novelty and excitement aspects of the ideas when they are reviewing.

	Overall	Novelty	Excitement	Feasibility	Effectiveness
Overall	–	0.725	0.854	0.097	0.642
Novelty	0.725	–	0.719	-0.073	0.357
Excitement	0.854	0.719	–	-0.031	0.565
Feasibility	0.097	-0.073	-0.031	–	0.251
Effectiveness	0.642	0.357	0.565	0.251	–

Table 10: Pairwise correlation between different metrics (symmetric matrix).

6.3 Reviewing Ideas is Inherently Subjective

Finally, we acknowledge that reviewing is inherently subjective, and reviewing based on ideas rather than executed papers might be even more subjective. We investigate this using inter-reviewer agreement. Specifically, we randomly split reviewers of each paper into half, use one half to rank the top and bottom 25% of all ideas, and then measure agreement with the held-out set of reviewers.¹¹ As shown in the first block of Table 11, reviewers have a relatively low agreement (56.1%) despite the fact that we have provided detailed explanations for each metric in our review form. As a baseline comparison, the NeurIPS 2021 reviewer consistency experiment found 66.0% accuracy using this reviewer agreement metric in the balanced setting (Beygelzimer et al., 2021, Lu et al., 2024). We also computed the reviewer agreement using the same metric on the 1.2K ICLR 2024 submissions related to language models, which has a balanced accuracy of 71.9%. While our reviewer agreement is higher than random (50%), it is generally lower than conference reviewing, most likely due to the higher subjectivity involved when evaluating ideas without seeing the actual experiment results.

7 Limitations of LLMs

With our findings from the human study in mind, we now turn to LLM performance to provide insights that could inform future methods for improving idea generation systems. Our ideation agent is motivated by two potential strengths of LLMs: their ability to scale by generating a vast number of ideas - far more than any human could - and the possibility of filtering these ideas to extract the best ones from the large pool. In theory, this approach could lead to high-quality ideas by leveraging inference scaling. However, we present empirical evidence that this naive assumption about scaling idea generation has significant limitations.

7.1 LLMs Lack Diversity in Idea Generation

We adopted an over-generate and rank paradigm in idea generation. This raises the question: is there an upper limit to how many new ideas LLMs can generate? To answer this question, we take a closer look at 4000 generated seed ideas for each topic.

We encode all raw ideas with `all-MiniLM-L6-v2` from Sentence-Transformers. For each idea, we compute its cosine similarity with all previously generated ideas on the same topic. We consider an idea as a duplicate if it has a similarity of above 0.8 with any of the previously generated ideas. In Figure 4, we show that as the agent keeps generating new batches of ideas, the percentage of non-duplicates in newly generated batches keeps decreasing, and the accumulated non-duplicate ideas eventually plateau. In fact, out of the 4000 generated seed ideas, there are only 200 non-duplicate

¹¹This metric follows the balanced accuracy metric as used in Lu et al. (2024) and avoids the limitations of other agreement metrics like Krippendorff’s alpha, which require overlapping reviews and would result in a sparse matrix due to the non-overlapping nature of our reviewer assignments. We do the random splitting 20 times and report the average to reduce variances.

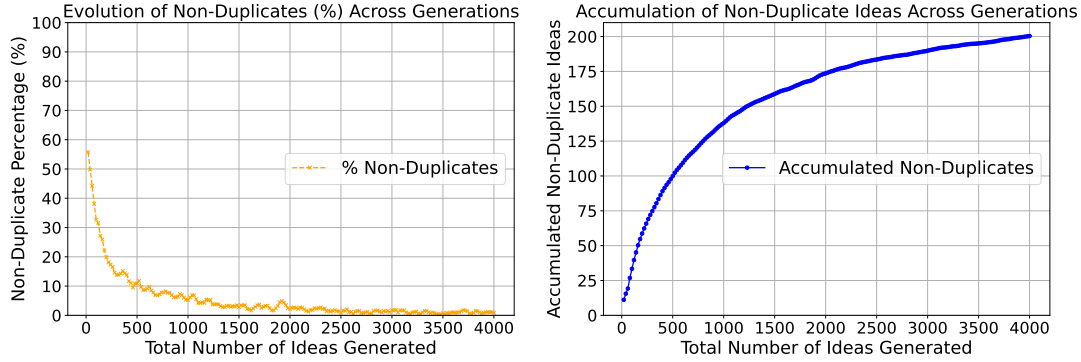


Figure 4: Measuring duplication of AI-generated ideas: the left figure plots the percentage of non-duplicate ideas in each new bucket of generated ideas; the right figure plots the accumulated non-duplicate ideas as the agent keeps generating new ideas. All data points are averaged across all topics.

unique ideas. This sets a bottleneck on our inference-time scaling since increasing the number of generated ideas simply leads to repeating duplicate ideas.

7.2 LLMs Cannot Evaluate Ideas Reliably

Most prior works have adopted *LLM-as-a-judge* for evaluating research ideas (Lu et al., 2024) motivated by the observation that LLMs can have a higher agreement with human evaluators than the inter-human agreement. However, we offer some empirical evidence that LLMs cannot evaluate ideas reliably yet.

Concretely, we use the average review score of each idea to rank the top and bottom 25% of all our collected human and AI ideas, and use this to benchmark various LLM evaluators. Specifically, we obtain the LLM predicted scores of all ideas and set the median score as the threshold to measure their accuracy on our balanced idea ranking data.

In the second block of Table 11, we compare several different LLM evaluators: 1) directly giving the review criteria and prompting for a final score (Baek et al., 2024, Li et al., 2024, Yang et al., 2024); 2) our pairwise ranker as described in Section 3.3; and 3) the “AI Scientist” reviewer agent (Lu et al., 2024). All of these LLM evaluators **have a lower agreement than our expert reviewers’ scores**. Even the best LLM evaluator — our own Claude-3.5 pairwise ranker — only achieves an accuracy of 53.3%, lower than our inter-reviewer consistency of 56.1%.

Even if AI-human agreement eventually matches or exceeds human-human agreement, simply meeting this baseline does not imply that AI-as-a-reviewer is meaningful, since we may be trading variance for bias, where AI reviewers are more consistent but rely on spurious correlations (Durmus et al., 2022). Our findings in Table 11 are consistent with these brittleness concerns, as we find a significant drop in AI-human agreement scores under our study compared to the original studies.

Finally, even though Claude-3.5 pairwise agreements may seem close to human agreement, many other pieces of evidence throughout the paper leads us to be cautious about the use of LLM-as-a-judge

	Consistency
Random	50.0
NeurIPS’21	66.0
ICLR’24	71.9
Ours	56.1
GPT-4o Direct	50.0
GPT-4o Pairwise	45.0
Claude-3.5 Direct	51.7
Claude-3.5 Pairwise	53.3
“AI Scientist” Reviewer	43.3

Table 11: Review score consistency among human reviewers (first block) and between humans and AI (second block).

in such a complex and subjective task. These include our findings on the significant discrepancy between the agent’s top-ranked ideas and the human expert’s top-ranked ideas (Appendix I) and how the AI Ideas + Human Rerank condition tends to score higher than the AI Ideas condition on all metrics in Section 5. These limitations of LLM auto-evaluation not only constrain the effectiveness of our over-generate-and-rank paradigm for idea generation but also raise concerns about trusting conclusions that are based primarily on LLM evaluators.

8 Qualitative Analysis and Examples

In this section, we offer some qualitative analysis of human- and AI-generated ideas based on our collected reviews and present four pairs of randomly sampled human and AI ideas as case studies.

8.1 Analysis of Free-Text Reviews

Following recent practices of using LLMs to extract patterns from text corpora (Zhong et al., 2022, 2023), we use Claude-3.5 to extract and cluster the main points from all reviews. We then manually verified and labeled each cluster.

Many reviews reinforce our quantitative finding that AI ideas tend to be more novel. For example, reviewers noted: “The idea of [...] is quite novel in an in-context learning setting.”, “The idea of exploring [...] using an LLM-based iterative approach is novel.”, “The idea of [...] when constructing prompts to improve cross-lingual transfer is one that I have not heard of before.”, “I like the idea to [...], and think it will be helpful for other researchers in the community.”, “Combining [...] is a unique way of attempting to preserve the gist of the information while likely losing specific identifiers.”, and “Safeguarding using [...] is clearly novel. Similar ideas have not been seen in the related work.”.

Next, we summarize some common failure modes of AI ideas:

1. **Being too vague on implementation details.** For example, one reviewer noted: “I’m not super clear on the details of this lattice and how the model will be prompted, so I’m not super sure how well the model will complete these subtasks and how well-suited this particular structure is to completing the overall task.” and another reviewer noted: ““For analyzing the effectiveness of the method, the proposal only provides a very ad-hoc + hand-wavey suggestion to compare responses across predefined questions.” In another case, the AI idea is criticized for not considering practical implementation details: “I think in each of the steps, there is something hard to execute. For example, in step Constellation Formation, how do we do the weighted sum?” Similarly, other reviews noted: “It’s unclear how CLIP is connected to the language model and how training a CLIP model would enable the LM to understand images.”, and “There’s no mentioning on how to prompt the model to generate defensive strategies and refine the model’s responses using these strategies.” Such vagueness often makes it difficult for reviewers to make confident judgments: “Because this idea is too general and vague, I can’t really answer the previous question. An idea needs a certain level of details to be determined if it fits for a conference/journal but this one misses them.”
2. **Misuse of datasets.** For example: “I’m not sure about the datasets picked. StereoSet is not a QA dataset; it simply contains statements. Also, I don’t understand why Dialogue NLI responses require empathy.”, “I’m concerned the datasets proposed are the right test cases for security of the code (since they are really just ML/programming problems, not system-level programming).”, and “the choice of datasets might not be the best to show the effect of incorporating multiple perspectives, especially TruthfulQA and ScienceQA, which seems to have a single correct interpretation and answer.” In another example, the benchmark datasets chosen are considered